

SRIE

Project 3 - Mediation under confidentiality constraints

BILEL HATMI, SUPERVISION

EIGHT WEEKS

4 August to 28 September
2026



The project

Deriving a game-theory model adapted to LLMs for solving conflicts under confidentiality constraints

Exposure or fear of judgement blocks conflict resolution more than any missing fix

An illustration: Ben and Anna

- Anna declines a high-visibility project and misses deadlines on the work she keeps. Ben misreads it, quietly filing her commitment as slipping.
- The real cause is a health episode she keeps from colleagues and manager alike, having watched a colleague who chose to speak up get filed as fragile — a label that rarely comes off.
- Two competing readings of the same six weeks, and the costlier one is the one that sticks.

A toll running to trillions of dollars

\$1T

- WHO and ILO put the yearly toll of **unresolved workplace conflict at roughly 12 billion lost working days**, or close to **\$1 trillion in forgone productivity**.¹

Effective, but rarely reached

- **Mediation resolves disputes** fully or substantially 74% of the time it is actually tried, against 44% across every other channel combined.²
- While mediation works when tried, **most disputes stall first**: 53% of people harassed at work never report the most recent incident, so the 74% excludes every stalled case.³

For Anna, as for most of that 53%, **the block sits in exposure to a human third party and the judgment that follows.**

The next page asks why a human mediator cannot close that exposure.

LLMs are better placed than humans to mediate under confidentiality constraints

Every route to a neutral human mediator closes

- Dziuda and Gradwohl concede the point themselves: the problem “**could be easily solved if an expert benevolent mediator were available,**” yet “**a mediator with expertise sufficient to judge the viability of the presented evidence is unlikely to be benevolent.**”¹
- Their objection lands here unchanged: Anna carries exactly what they call “evidence.”
- A human mediator rarely arrives without a stake — a tie to one of the parties, a rung on the org chart near one of them, or pressure from the paying employer. Anyone can shape what gets heard, whether or not the mediator notices.



A model's function, not the mediator's promise

- This project makes **neutrality a property of the objective function**, with **collective welfare inside it** as a measurable quantity to be optimised, so any bias would show up in what the model is built to maximise.

Two consequences of the shift

- **The fear of judgement recedes:** no third person is in the room to form a view, so disclosure becomes franker and more accurate.
- **Verbal register becomes a lever of persuasion and noising:** framing can build empathy without ever naming the sensitive fact, and a persona — counsellor, assessor, plain negotiator — can be chosen for the party being addressed.

An intention can be declared and broken, whereas **a function can optimise collective well-being by design.**

The next page turns Anna and Ben into a formal two-player problem.

Maximising resolution **without** spending confidentiality

Our illustration

- Anna holds the only sensitive fact in the room; her manager Ben can act on a belief, right or wrong.
- Information is what turns Ben’s belief, yet Anna cannot move it without giving up the fact.
- The mediator therefore has two jobs:
 - read the facts as they stand;
 - move Ben’s belief towards collective well-being without exposing Anna.

An optimisation problem under two hard constraints

Objective: a mediator **maximises how far the conflict resolves**.

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- **Confidentiality** binds that objective for both parties at once.
 - No manipulation: **neither party can exploit the other’s moves or data**.



Three pieces of work answer that objective

- 1 **Model the situation with game theory**, in a structure an LLM can run autonomously — carrying existing work on human mediators over to an LLM, and making it practical.
- 2 **Model the agentic workflow**: measuring, adapting and optimising strategy, and assessing confidentiality at each step.
- 3 **Implement worked examples as a proof of concept**: synthetic for now, with personas standing in for real recruits.

Confidentiality and resolution are optimised together.

The closing page returns to what this changes for Anna and Ben.

A mediator does not replace the relationship but **keeps it alive**

Inverting the common assumption

The common assumption underneath all of this is that an AI tool has no business in human relationships.

- Here it is used **to correct a failure mode of human relationships**, above all where one party holds power over the other.

A deliberate two-player scope

- **Two players is the chosen frame**, and the argument holds unchanged as the game scales up.



Bounded here, but built to generalise

- 1 **Reliable work here reaches past mediation.**
- 2 **Structures designed for trust and safety** give a reward function worth maximising.
- 3 **Such structures scale**, as long as the properties they model continue to hold in reality.

Anna's actual choice was never between disclosure and silence; it was between a system that **reads exposure as risk** and one that does not.



Supervising the students

This program is meant to be more than just a research project

More than a project

Four things get built alongside the research, and they are on the timetable like everything else.

Method



- **3 cross-cutting themes**, everyone in the room regardless of project: **literature review, reading and writing a paper, designing an experiment.**
- Turning a qualitative question into **a valid statistical design**, and **criticising an experiment while reading it.**

Tools



- **Claude Code** across the whole range, **from literature to code, wired into a single connected ecosystem.**
- Using **organisational tools like Jira, Drive and Slack** to improve efficiency and communication

Communication



- **Presenting technical work** in a structured way
- **Asking the right questions** at the right moment **to encourage creativity**, and reacting appropriately when knowledge is missing.

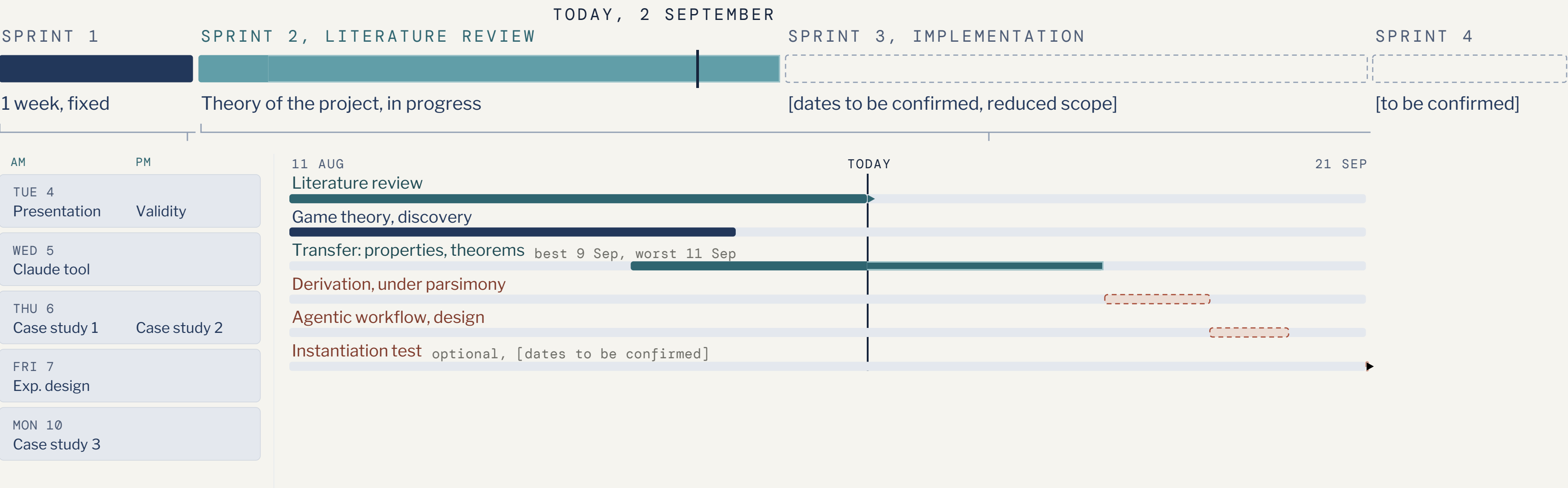
Yourself



- **A time to get to know yourself and work more efficiently** (Productivity, Rest, Learning, feedback culture, fear of comparison, perfectionism...). These are pain points common to highly demanding work such as research.

The team is measuring what AI does to the people who use it, so it is unusually well placed **to study its own use of AI tools.**
The next page sets out the eight-week calendar.

The roadmap



The experimental scope might be reduced, the theoretical phasis being more complex than anticipated

How the stream runs

Three levels, each carrying a different register of work. No level does the job of another.



The whole cohort for **what is deemed useful to everyone**, the group meeting to **keep the needs and ambitions of each stream coherent**, the individual meeting **to delve deeper into technical issues**.

Core bibliography for now

1. The reasons a mediator work

Implementation of Communication Equilibria by Correlated Cheap Talk:
The Two-Player Case.

Vida and Forges, 2013. Theoretical Economics, 8(1)

2. The formal mechanism of noise and register

Mediation, Arbitration and Negotiation.

Goltsman et al., 2009. Journal of Economic Theory, 144(4)

The Logic of Indirect Speech.

Pinker, Nowak and Lee, 2008. PNAS, 105(3)

Coarse Thinking and Persuasion.

Mullainathan et al., 2008. Quarterly Journal of Economics, 123(2)

3. The empirical ground, and the substrate

What Happens in Vagueness.

Massenkoff et al., 2025. Journal of Economic Behavior and Organization, 239

KODIS: A Multicultural Dispute Resolution Dialogue Corpus.

Hale et al., 2025. arXiv:2504.12723

4. The system targeted, and the probe

Truthful AI Advisors: A Pre-Specified Benchmark for Large Language Model
Honesty Under Preference Misalignment.

Balyani et al., 2026. arXiv:2606.01456

Beyond Memorization: Violating Privacy via Inference with Large
Language Models.

Staab et al., 2024. ICLR. arXiv:2310.07298